Unit 2 Classwork — Scientific Notation — Answer Key

Scientific Notation

A. Scientific Notation — Conversions

Problem 1 Score 3

Write 32,000 in scientific notation.

Answer: 3.2×10^{4}

Problem 2 Score 3

Write 0.0091 in scientific notation.

Answer: 9.1×10^{-3}

Problem 3 Score 3

Write 5.6×10^{6} in standard form.

Answer: 5,600,000

Problem 4 Score 4

Write 2.3×10^{-4} in standard form.

Answer: 0.00023

Problem 5 Score 4

Write 470,000,000 in scientific notation.

Answer: 4.7×10^{8}

B. Scientific Notation — Operations

Problem 1 Score 4

Multiply: $(2 \times 10^3) \times (3 \times 10^5)$

Answer: 6×10^8

Problem 2 Score 4

Divide: $(8 \times 10^7) \div (2 \times 10^3)$

Answer: 4×10^4

Problem 3 Score 5

Multiply: $(6 \times 10^4) \times (5 \times 10^2)$ — adjust if needed!

Answer: 3×10^7

Problem 4 Score 5

A server stores 4.5×10^9 photos. Each photo uses 2×10^6 bytes. How many bytes in all? (scientific notation)

Answer: 9×10^{15}

Problem 5 Score 5

Light travels 3×10^8 meters each second. There are 3.6×10^3 seconds in an hour. How many meters does light travel in 2 hours?

A. 21.6×10^{11}

B. 2.16×10^{12}

C. 1.08×10^{12}

D. 2.16×10^{24}

Answer: B

Problem 6 Score 5

A city has 8×10^5 people. Each person uses about 3×10^2 liters of water a day. Sam says the city uses 2.4×10^3 liters a day. Is that reasonable?

A. No. $8 \times 3 = 24$ and $10^5 \times 10^2 = 10^7$, so it is 2.4×10^8 liters.

B. Yes. $5 - 2 = 3$, so the exponent is 3.

C. No. It is 24×10^7 liters, and that is the final answer.

D. Yes. One person uses 300 liters, and 2.4×10^3 is more than 300.

Answer: A